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Inventor(s)

TANAKA; Hiroyuki

PRESS-FORMING METHOD AND METHOD OF MANUFACTURING PRESS-FORMED PRODUCT

Abstract

A press-forming method includes: forming a blank having a concave shape corresponding to a curve of a concave curved portion of a target shape; press-forming the formed blank into a press-formed product having an intermediate shape having a top portion, a vertical wall portion continuous from the top portion, a flange portion continuous from the vertical wall portion, a concave curved portion curved in a concave shape in a top view, and a cutout portion located at a part of the flange portion and the vertical wall portion or at a part of the flange portion, the vertical wall portion, and the top portion in a central portion of the curve of the concave curved portion; and press-forming the press-formed product having the intermediate shape into the target shape. The press-forming is performed such that a part of the concave shape provided in the blank becomes the cutout portion.

Inventors:	TANAKA; Hiroyuki (Tokyo, JP)
Applicant:	JFE STEEL CORPORATION (Tokyo, JP)
Family ID:	89026541
Assignee:	JFE STEEL CORPORATION (Tokyo, JP)
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Background/Summary

FIELD

[0001] The present invention relates to a press-forming method and a method of manufacturing a press-formed product of press-forming, as a target shape, a press-formed product having a top portion and a vertical wall portion continuous from the top portion and including a concave curved portion curved in a concave shape in a top view.

BACKGROUND

[0002] In automobiles and home electric appliances, many press-formed products have been used as component members, and the press-formed products have been manufactured by press-forming a blank that is a metal plate. In recent years, a high-strength metal plate such as a high-tensile steel plate or an aluminum alloy plate having a tensile strength of 590 MPa or more has been used for press-forming. However, since such a metal plate has low ductility and a low Lankford value, there has been a problem in that cracking is likely to occur during press-forming.

[0003] As an example of a shape of the press-formed product in which cracking is likely to occur, there is a press-formed product having a top portion and a vertical wall portion continuous from the top portion with a ridge portion interposed therebetween and including a concave curved portion in which the top portion, the ridge portion, and the vertical wall portion are curved in a concave shape in top view. When such a press-formed product is press-formed from a blank that is a metal plate, cracking (stretch flange cracking) is likely to occur due to tensile deformation in a direction along a curve of the blank in a press-forming process, which causes a forming defect.

[0004] Therefore, conventionally, a technique for suppressing occurrence of stretch flange cracking in the press-forming process of the press-formed product having the concave curved portion in a top view has been proposed.

[0005] For example, Patent Literature 1 discloses a technique for suppressing occurrence of stretch flange cracking in a flange portion when a sheet metal formed product including a top portion having a concave outer peripheral edge portion and the flange portion (corresponding to a vertical wall portion of the present application) continuously curved from the concave outer peripheral edge portion is press-formed. In such technique, first, the flange portion including large curvature portions having a large curvature and a small curvature portion having a small curvature interposed between the large curvature portions is formed on a blank to obtain a pre-formed product. The curvature of the large curvature portion is larger than that of a curvature portion included in the flange portion of the sheet metal formed product, and the curvature of the small curvature portion is smaller than that of the curvature portion included in the flange portion of the sheet metal formed product. Further, a line length of the flange portion of the pre-formed product in a plan view is made shorter than a line length of the flange portion of the sheet metal formed product in the plan view. The cracking of the flange portion can be prevented by re-striking the pre-formed product obtained as such into a target shape.

[0006] In addition, Patent Literature 2 discloses a technique in which a top portion and a vertical wall portion that is connected to the top portion via a bent portion having an arc-shaped curved

portion and has a flange portion disposed on the opposite side of the bent portion are provided, and a component including the top portion, the vertical wall portion, and the flange portion is formed from a raw metal plate. In such technique, the vertical wall portion and the flange portion are formed as an end portion of a portion corresponding to the lower side of the L shape of the raw metal plate slides on a portion corresponding to a top portion of a die tool of press forming while at least a part of a portion corresponding to a top portion of the raw metal plate is pressurized by a pad. As a result, cracking of the flange portion can be suppressed.

CITATION LIST

Patent Literature

[0007] Patent Literature 1: JP 2021-159951 A

[0008] Patent Literature 2: JP 5168429 B2

SUMMARY

Technical Problem

[0009] However, according to the technique disclosed in Patent Literature 1, when the press-formed product having the top portion and the vertical wall portion continuous from the top portion with the ridge portion interposed therebetween and including the concave curved portion that is curved in the concave shape in the top view is press-formed, stretch flange cracking is likely to occur in some cases. In particular, when the radius of curvature of the concave curved portion is small (the curvature is large), a line length of a vertical wall portion of a press-formed product having an intermediate shape (the flange portion of the pre-formed product of Patent Literature 1) is too short to sufficiently reduce tensile strain concentrated on the vertical wall portion of the concave curved portion. As a result, since an end portion of the vertical wall portion at the concave curved portion is formed by being subjected to large tensile deformation in a direction along the curve, there is a problem in that tensile strain is concentrated and, as such, stretch flange cracking is likely to occur.

[0010] Further, in the technique disclosed in Patent Literature 2, the vertical wall portion and the flange portion are formed while the end portion of the portion corresponding to the lower side of the L shape of the raw metal plate slides on the portion corresponding to the top portion of the die tool of press forming, so that the raw metal plate is moved during press-forming. As a result, there is a problem in that a change in the shape of a press-formed component increases, dimensional accuracy fluctuates, and an effect of suppressing stretch flange cracking also fluctuates.

[0011] The present invention has been made to solve the above-described problems, and an object of the present invention is to provide a press-forming method and a method of manufacturing a press-formed product capable of sufficiently suppressing stretch flange cracking of the press-formed product having a top portion and a vertical wall portion and including a concave curved portion that is curved in a concave shape in a top view.

Solution to Problem

[0012] To solve the above-described problems and achieve the object, [0013] (1) A press-forming method according to the present invention press-forms a press-formed product into a target shape, the target shape including a top portion, a vertical wall portion continuous from the top portion, a flange portion continuous from the vertical wall portion, and a concave curved portion curved in a concave shape in a top view, and includes: a blank forming step of forming a blank provided with a concave shape corresponding to a curve of the concave curved portion of the target shape; an intermediate shape forming step of press-forming the formed blank into a press-formed product having an intermediate shape, the intermediate shape having a top portion, a vertical wall portion continuous from the top portion, a flange portion continuous from the vertical wall portion, a concave curved portion curved in a concave shape in a top view, and a cutout portion located at a part of the flange portion and the vertical wall portion or at a part of the flange portion, the vertical wall portion, and the top portion in a central portion of the curve of the concave curved portion; and a target shape forming step of press-forming the press-formed product having the intermediate

shape into the target shape, wherein the press-forming is performed in the intermediate shape forming step such that a part of the concave shape provided in the blank becomes the cutout portion. [0014] (2) A press-forming method according to the present invention press-forms a press-formed product into a target shape, the target shape including a top portion, a vertical wall portion continuous from the top portion, and a concave curved portion curved in a concave shape in a top view, and includes: a blank forming step of forming a blank provided with a concave shape corresponding to a curve of the concave curved portion; an intermediate shape forming step of press-forming the formed blank into a press-formed product having an intermediate shape, the intermediate shape including a top portion, a vertical wall portion continuous from the top portion, a concave curved portion curved in a concave shape in a top view, and a cutout portion located at a part of the vertical wall portion and the top portion in a central portion of the curve of the concave curved portion; and a target shape forming step of press-forming the press-formed product having the intermediate shape into the target shape, wherein press-forming is performed in the intermediate shape forming step such that a part of the concave shape provided in the blank becomes the cutout portion. [0015] (3) In the above-described press-forming method according to (1) or (2), in the intermediate shape forming step, a radius of curvature RE at an end portion of the curve of the concave curved portion having the intermediate shape is set to be larger than a radius of curvature $R_{sub.0}$ at the concave curved portion having the target shape. [0016] (4) A method according to the present invention of manufacturing a press-formed product having a target shape, the target shape including a top portion, a vertical wall portion continuous from the top portion, a flange portion continuous from the vertical wall portion, and a concave curved portion curved in a concave shape in a top view, includes: a blank forming step of forming a blank provided with a concave shape corresponding to a curve of the concave curved portion of the target shape; an intermediate shape forming step of press-forming the formed blank into a press-formed product having an intermediate shape, the intermediate shape including a top portion, a vertical wall portion continuous from the top portion, a flange portion continuous from the vertical wall portion, a concave curved portion curved in a concave shape in a top view, and a cutout portion located at a part of the flange portion and the vertical wall portion or at a part of the flange portion, the vertical wall portion, and the top portion in a central portion of the curve of the concave curved portion; and a target shape forming step of press-forming the press-formed product having the intermediate shape into the target shape, wherein press-forming is performed in the intermediate shape forming step such that a part of the concave shape provided in the blank becomes the cutout portion. [0017] (5) A method according to the present invention of manufacturing a press-formed product having a target shape, the target shape including a top portion, a vertical wall portion continuous from the top portion, and a concave curved portion curved in a concave shape in a top view, includes: a blank forming step of forming a blank provided with a concave shape corresponding to a curve of the concave curved portion; an intermediate shape forming step of press-forming the formed blank into a press-formed product having an intermediate shape, the intermediate shape including a top portion, a vertical wall portion continuous from the top portion, a concave curved portion curved in a concave shape in a top view, and a cutout portion located at a part of the vertical wall portion and the top portion in a central portion of the curve of the concave curved portion; and a target shape forming step of press-forming the press-formed product having the intermediate shape into the target shape, wherein press-forming is performed in the intermediate shape forming step such that a part of the concave shape provided in the blank becomes the cutout portion. [0018] (6) In the above-described method of manufacturing a press-formed product according to (4) or (5), the press-forming is performed in the intermediate shape forming step such that a radius of curvature $R_{sub.E}$ at an end portion of the curve of the concave curved portion having the intermediate shape is set to be larger than a radius of curvature R_o at the concave curved portion having the target shape.

Advantageous Effects of Invention

[0019] In a press-forming method and a method of manufacturing a press-formed product according to the present invention, a blank provided with a concave shape corresponding to a curve of a concave curved portion is formed, and the formed blank is press-formed into a press-formed product having an intermediate shape provided with a cutout portion at a central portion of the curve of the concave curved portion. Here, press-forming is performed so that the concave shape provided in the blank becomes the cutout portion. Accordingly, it is possible to reduce tensile stress acting on the vicinity of the cutout portion and to disperse the tensile stress on both end sides of the curve at the concave curved portion. As a result, when the press-formed product having the intermediate shape is press-formed into a target shape, it is possible to sufficiently suppress stretch flange cracking, to enable stable press-forming without cracking, and to improve productivity and yield.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0020] FIG. 1 is a flowchart illustrating a flow of processing in a press-forming method according to an embodiment.

[0021] FIG. 2 is a perspective view illustrating an example of a press-formed product to be formed in the present embodiment ((a) Z-shaped cross-sectional shape, (b) backward C-shaped cross-sectional shape, (c) hat cross-sectional shape).

[0022] FIG. 3 is a diagram illustrating a conventional technique of press-forming a press-formed product having a concave curved portion in a top view in one step.

[0023] FIG. 4 is a contour diagram illustrating an example of a result of a plate thickness reduction rate of the press-formed product press that was formed in one step according to the conventional technique.

[0024] FIG. 5 is a diagram illustrating a pre-formed product (a press-formed product having an intermediate shape) to be press-formed by a method under comparative conditions.

[0025] FIG. 6 is a contour diagram illustrating an example of a result of a plate thickness reduction rate of each of the pre-formed product (the press-formed product having the intermediate shape) and a sheet metal formed product (a press-formed product having a target shape) that were press-formed by the method under comparative conditions.

[0026] FIG. 7 is a diagram illustrating the press-formed products having the intermediate shape and the target shape that are press-formed by the press-forming method according to the embodiment.

[0027] FIG. 8 is a diagram illustrating a step of press-forming the press-formed product having the concave curved portion in the top view by the press-forming method according to the embodiment ((a) intermediate shape forming step, (b) target shape forming step).

[0028] FIG. 9 is a contour diagram illustrating an example of the result of the plate thickness reduction rate of the press-formed products respectively having the intermediate shape and the target shape that are press-formed by the press-forming method according to the embodiment.

[0029] FIG. 10 is a diagram illustrating a suitable position of a ridge portion at a concave curved portion of the press-formed product having the intermediate shape to be press-formed in the press-forming method according to the present invention.

[0030] FIG. 11 is a diagram illustrating a specific example of a cutout portion at the concave curved portion of the press-formed product having the intermediate shape to be press-formed in the press-forming method according to the present invention.

[0031] FIG. 12 is a diagram illustrating a target shape of a press-formed product having a hat cross-sectional shape to be formed in an example.

[0032] FIG. 13 is a diagram illustrating a step of press-forming each of the press-formed products having the intermediate shape and the target shape by the method according to the present

invention in the example ((a) intermediate shape forming step, (b) target shape forming step).
[0033] FIG. **14** is a diagram illustrating a conventional example in which a press-formed product having a target shape is press-formed in one step in the example.
[0034] FIG. **15** is a contour diagram of a plate thickness reduction rate obtained for the press-formed product having the target shape in the conventional example in the example.
[0035] FIG. **16** is a contour diagram of a plate thickness reduction rate obtained for press-formed products having an intermediate shape and a target shape in a comparative example in the example ((a) press-formed product having intermediate shape, (b) press-formed product having target shape).
[0036] FIG. **17** is a contour diagram of a plate thickness reduction rate obtained for press-formed products having an intermediate shape and a target shape in a first invention example in the example ((a) press-formed product having intermediate shape, (b) press-formed product having target shape).
[0037] FIG. **18** is a contour diagram of a plate thickness reduction rate obtained for press-formed products having an intermediate shape and a target shape in a second invention example in the example ((a) press-formed product having intermediate shape, (b) press-formed product having target shape).

DESCRIPTION OF EMBODIMENTS

[0038] Hereinafter, embodiments of a press-forming method and a method of manufacturing a press-formed product according to the present invention will be described. Note that the present invention is not limited by the present embodiment.
[0039] Prior to describing the press-forming method and the method of manufacturing the press-formed product according to the embodiment of the present invention, a press-formed product to be formed in the present invention and circumstances leading to the present invention will be described.

Press-Formed Product

[0040] FIG. **2** is a perspective view illustrating a specific example of a press-formed product to be formed in the present invention.
[0041] A press-formed product **210** in FIG. **2(a)** is an example of a press-formed product having a Z-shaped cross-sectional shape and including a top portion **211**, a vertical wall portion **215** continuous from the top portion **211** with a ridge portion **213** interposed therebetween, and a flange portion **219** continuous from the vertical wall portion **215** with a die shoulder portion **217** interposed therebetween. The press-formed product **210** includes a concave curved portion **221** in which the ridge portion **213**, the vertical wall portion **215**, the die shoulder portion **217**, and the flange portion **219** side are curved in a concave shape in a top view, and linear portions **223** respectively extending in a linear shape from both ends of the curve of the concave curved portion **221**.

[0042] Further, a press-formed product **230** in FIG. **2(b)** is an example of a press-formed product having a backward C-shaped cross-sectional shape and including a top portion **231** and vertical wall portions **235** respectively continuous from both end sides of the top portion **231** with ridge portions **233** interposed therebetween. The press-formed product **230** includes a concave curved portion **241** in which a ridge portion **233a** and a vertical wall portion **235a** side are curved in the concave shape in the top view, and linear portions **243** respectively extending in the linear shape from both ends of the curve of the concave curved portion **241**.

[0043] Further, a press-formed product **250** in FIG. **2(c)** is an example of a press-formed product having a hat cross-sectional shape and including a top portion **251**, vertical wall portions **255** respectively continuous from both end sides of the top portion **251** with ridge portions **253** interposed therebetween, and flange portions **259** continuous from the vertical wall portion **255** with die shoulder portions **257** interposed therebetween. The press-formed product **250** includes a concave curved portion **261** in which a ridge portion **253a**, a vertical wall portion **255a**, a die

shoulder portion **257a**, and a flange portion **259a** side are curved in the concave shape in the top view, and linear portions **263** respectively extending in the linear shape from both ends of the curve of the concave curved portion **261**.

[0044] As such, the press-formed product to be formed in the present invention may be any press-formed product as long as the press-formed product includes a top portion and a vertical wall portion continuous from the top portion with a ridge portion interposed therebetween and includes a concave curved portion that is curved in the concave shape in the top view as illustrated in FIG.

2. Therefore, the press-formed product to be formed does not limit other specific shapes, for example, a cross-sectional shape, a radius of curvature of the curve of the concave curved portion, a length of the linear portion, and the like.

[0045] Hereinafter, a case in which the press-formed product **210** having the Z-shaped cross-sectional shape illustrated in FIG. **2(a)** is press-formed as a target shape will be described as an example.

Circumstances Leading to Present Invention

[0046] Conventionally, the press-formed product **210** having the Z-shaped cross-sectional shape including the concave curved portion **221** in the top view as illustrated in FIG. **2(a)** is press-formed into a target shape in one step using a tool of press forming **21** including a die **23**, a punch **25**, and a pad **27** as illustrated in FIG. **3**. Here, the concave curved portion **221** formed by a die side concave curved portion forming portion **23a** and a punch side concave curved portion forming portion **25a** is subjected to stretch flange forming.

[0047] For the press-formed product **210** that is press-formed into the target shape in one step, the inventors of the present invention performed finite element method analysis (FEM analysis) in the process of press-forming using the die **21** illustrated in FIG. **3** and determined a plate thickness reduction rate of the press-formed product **210** after the press-forming. Here, the plate thickness reduction rate is a value obtained by subtracting a plate thickness of each portion after press-forming from a plate thickness of a blank **100**, that is a metal plate before press-forming, and dividing the obtained value by the plate thickness of the blank **100**.

[0048] As illustrated in FIG. **4**, in the press-formed product **210** that is press-formed in one step conventionally, the plate thickness reduction rate of the end portion of the flange portion **219** at the concave curved portion **221** is 21.5%, that is the largest value, and it can be seen that cracking is likely to occur due to stretch flange forming.

[0049] To suppress the occurrence of cracking due to such stretch flange forming, Patent Literature 1 described above proposes a method of performing press-forming in two steps including a flange-up step (pre-forming step) and a re-strike step.

[0050] In the method of Patent Literature 1, first, in the flange-up step, as illustrated in FIG. **5**, a press-formed product **170** having an intermediate shape that is a pre-formed product is obtained. Then, in the subsequent re-strike step, the pre-formed product is re-struck into the press-formed product **210** having the target shape that is the sheet metal formed product.

[0051] In the method of Patent Literature 1, the press-formed product **170** having the intermediate shape includes two end portions **181b** of the curve having a large curvature (small radius of curvature $R_{sub.E}$) and a central portion **181a** of the curve having a small curvature (large radius of curvature $R_{sub.C}$) and interposed between the end portions **181b** of the curve.

[0052] Here, as the press-formed product **170** having the intermediate shape, a condition is considered in which a ridge length (curve efgh) of a curve at a concave curved portion **181** is shorter than a ridge length (curve pq) of the curve at the concave curved portion **221** of the press-formed product **210** having the target shape illustrated in FIG. **5**. Such condition satisfies a condition of "a line length of the flange portion of the pre-formed product in a plan view is made shorter than a line length of the flange portion of the sheet metal formed product in the plan view." disclosed in Patent Literature **1**. Here, the ridge length (curve efgh) of the curve at the concave curved portion **181** of the press-formed product **170** having the intermediate shape refers to a

length from a starting point e to an end point h in the direction along the curve of a ridge portion **173**. The ridge length (curve pq) of the curve at the concave curved portion **221** of the press-formed product **210** having the target shape refers to a length from a start point p to an end point q in the direction along the curve of the ridge portion **213** indicated by a broken line.

[0053] Then, the inventors of the present application performed finite element method analysis for each step under the conditions illustrated in FIG. 5 to determine the plate thickness reduction rate of each of the press-formed product **170** having the intermediate shape and the press-formed product **210** having the target shape.

[0054] As illustrated in FIG. 6, the plate thickness reduction rate of the press-formed product **170** having the intermediate shape under the conditions illustrated in FIG. 5 was 24.4% at the maximum. Further, the plate thickness reduction rate of the press-formed product **210** having the target shape was 25.4% at the maximum. As such, under the conditions illustrated in FIG. 5, the plate thickness reduction rate was increased as compared with the case of press-forming in one step conventionally (FIG. 4), leading to frequent occurrence of stretch flange cracking. That is, it was found that the condition illustrated in FIG. 5 had almost no effect of suppressing stretch flange cracking.

[0055] Therefore, the inventors of the present application have intensively studied means for suppressing such stretch flange cracking. In the study, the inventors of the present application have focused on a portion at which stretch flange cracking occurs in a conventional method. That is, in the press-formed product **210** press-formed into the target shape by the conventional method, as illustrated in FIGS. 4 and 6, the plate thickness reduction rate is large at a central portion **221a** of the curve of the flange portion **219** of the concave curved portion **221** or in the vicinity thereof. Therefore, attention has been paid to the fact that stretch flange cracking is likely to occur at the aforementioned portion.

[0056] Therefore, the inventors of the present application have intensively studied a method of reducing tensile stress acting on the concave curved portion or the vicinity thereof and dispersing the tensile stress on both end sides of the curve at the concave curved portion in the stretch flange forming in which the tensile stress acts on the concave curved portion. As a result, it has been conceived that, by forming a part of a concave curved portion in a notched shape in a press-formed product having an intermediate shape, tensile stress acting on the concave curved portion or the vicinity thereof is reduced and the tensile stress on both end sides of the curve at the concave curved portion is dispersed. It has been found that it is possible to suppress stretch flange cracking in the press-formed products respectively having the intermediate shape and the target shape by reducing the tensile stress at the concave curved portion.

[0057] The present invention has been made based on such examination results, and the configuration thereof is described as follows.

Press-Forming Method

[0058] In the press-forming method according to the embodiment, the press-formed product **210** including the concave curved portion **221** that is curved in the concave shape in the top view as illustrated in FIG. 2(a) described above as an example is press-formed into a target shape. As illustrated in FIG. 1, the press-forming method includes a blank forming Step S1, an intermediate shape forming Step S3, and a target shape forming Step S5. Since the press-formed product **210** is manufactured by executing the above-described press-forming method, the invention of the press-forming method can be configured as the invention of the method of manufacturing the press-formed product. Therefore, the embodiment of the press-forming method described below is common to the embodiment of the method of manufacturing the press-formed product. Each of the above steps will be described below with reference to FIGS. 7 and 8.

<<Blank Forming Step>>

[0059] The blank forming Step S1 is a step of forming the blank **100** (refer to FIG. 8(a)) provided with a concave shape **101** corresponding to the curve of the concave curved portion **221** of the

press-formed product **210** having the target shape. The curve of the concave curved portion **221** refers to, for example, the curve of the ridge portion **213** at the concave curved portion **221**.

<<Intermediate Shape Forming Step>>

[0060] The intermediate shape forming Step S3 is a step of press-forming the blank **100** into a press-formed product **110** having an intermediate shape illustrated in FIG. 7(a). Here, the press-formed product **110** having the intermediate shape includes a top portion **111**, a vertical wall portion **115** continuous from the top portion **111** via a ridge portion **113** interposed therebetween, and a flange portion **119** continuous from the vertical wall portion **115** with a die shoulder portion **117** interposed therebetween. In addition, the press-formed product **110** having the intermediate shape includes a concave curved portion **121** that is curved in the concave shape in the top view and linear portions **123** that respectively extend in the linear shape from both ends of the curve of the concave curved portion **121**. Further, a cutout portion **125** having a shape in which the flange portion **119** and the vertical wall portion **115** are partially cut out is provided at a central portion **121a** of the curve of the concave curved portion **121**. Then, in the intermediate shape forming Step S3, press-forming is performed so that a part of the concave shape **101** provided in the blank **100** becomes the cutout portion **125**. Here, the cutout portion **125** provided at the vertical wall portion **115** of the concave curved portion **121** is a portion that should originally be the vertical wall portion **115** and a portion without any material. A length of the cutout portion **125** in a height direction is shorter than a length of the vertical wall portion **115** in the height direction in a portion other than the concave curved portion **121**.

[0061] In the intermediate shape forming Step S3, for example, as illustrated in FIG. 8(a), an intermediate shape forming tool of press forming **1** including an intermediate shape forming die **3**, an intermediate shape forming punch **5**, and a pad **7** can be used to press-forming the press-formed product **110** having the intermediate shape.

[0062] The intermediate shape forming die **3** includes a die side intermediate shape concave curved portion forming portion **3a** that forms the concave curved portion **121** having the intermediate shape and a die side intermediate shape linear portion forming portion **3b** that forms the linear portion **123** having the intermediate shape.

[0063] The intermediate shape forming punch **5** includes a punch side intermediate shape concave curved portion forming portion **5a** and a punch side intermediate shape linear portion forming portion **5b**. The punch side intermediate shape concave curved portion forming portion **5a** forms the concave curved portion **121** having the intermediate shape in cooperation with the die side intermediate shape concave curved portion forming portion **3a**. The punch side intermediate shape linear portion forming portion **5b** forms the linear portion **123** having the intermediate shape in cooperation with the die side intermediate shape linear portion forming portion **3b**.

[0064] The pad **7** presses the blank **100** toward the intermediate shape forming punch **5** to press the blank **100**. Note that, in the intermediate shape forming Step S3, the intermediate shape forming tool of press forming **1** does not necessarily include the pad **7**, and the intermediate shape forming die **3** may include a top plate forming portion.

<<Target Shape Forming Step>>

[0065] The target shape forming Step S5 is a step of press-forming the press-formed product **110** having the intermediate shape that was press-formed in the intermediate shape forming Step S3 into the press-formed product **210** having the target shape illustrated in FIG. 7(b).

[0066] In the target shape forming Step S5, for example, as illustrated in FIG. 8(b), press-forming can be performed using a target shape forming tool of press forming **11** including a target shape forming die **13**, a target shape forming punch **15**, and a pad **17**.

[0067] The target shape forming die **13** includes a die side target shape concave curved portion forming portion **13a** that forms the concave curved portion **221** having the target shape and a die side target shape linear portion forming portion **13b** that forms the linear portion **223** having the target shape.

[0068] The target shape forming punch **15** includes a punch side target shape concave curved portion forming portion **15a** and a punch side target shape linear portion forming portion **15b**. The punch side target shape concave curved portion forming portion **15a** forms the concave curved portion **221** having the target shape in cooperation with the die side target shape concave curved portion forming portion **13a**. The punch side target shape linear portion forming portion **15b** forms the linear portion **223** having the target shape in cooperation with the die side target shape linear portion forming portion **13b**.

[0069] The pad **17** presses the top portion **111** of the press-formed product **110** having the intermediate shape toward the target shape forming punch **15** to press the press-formed product **110**. Note that, in the target shape forming Step S5, the target shape forming tool of press forming does not necessarily include the pad **17**, and the target shape forming die **13** may include a top plate forming portion.

Operation Effects

[0070] In the press-forming method according to the embodiment, first, in the blank forming Step S1, the blank **100** provided with the concave shape **101** corresponding to the curve of the concave curved portion **221** of the press-formed product **210** having the target shape is formed. Next, in the press-forming method according to the embodiment, in the intermediate shape forming Step S3, the formed blank **100** is press-formed into the press-formed product **110** having the intermediate shape in which the cutout portion **125** is provided at the central portion **121a** of the curve of the concave curved portion **121**.

[0071] In the press-formed product **110** having the intermediate shape press-formed as such, tensile stress acting on the vicinity of the cutout portion **125** becomes small. Further, the tensile stress is dispersed on both end sides of the curve at the concave curved portion **121**. As a result, in the process of press-forming the press-formed product **110** having the intermediate shape into the press-formed product **210** having the target shape, it is possible to suppress stretch flange cracking and to press-forming the press-formed product having the target shape.

[0072] This is verified by finite element method analysis (FEM analysis) of the press-forming process in the press-forming method according to the embodiment illustrated in FIG. 8 as an example. In the verification, the finite element method analysis was performed for each of the process of press-forming the blank **100** provided with the concave shape **101** into the press-formed product **110** having the intermediate shape and the process of press-forming the press-formed product **110** having the intermediate shape into the press-formed product **210** having the target shape. Then, the plate thickness reduction rate was calculated for each of the press-formed product **110** having the intermediate shape and the press-formed product **210** having the target shape.

[0073] FIG. 9 is a contour diagram of the plate thickness reduction rate of each of the press-formed product **110** having the intermediate shape and the press-formed product **210** having the target shape. Here, in the press-formed product **110** having the intermediate shape, as illustrated in FIG. 10, the ridge portion **113** of the concave curved portion **121** having the intermediate shape is set to be on an outer side of the ridge portion **213** of the concave curved portion **221** having the target shape. The outer side of the ridge portion **213** having the target shape refers to a center side of the curve with respect to the curve pq at the concave curved portion **221** having the target shape. Further, the ridge length of the ridge portion **113** at the concave curved portion **121** (for example, 300 mm) is longer than the ridge length of the ridge portion **213** at the concave curved portion **221** having the target shape (for example, 250 mm).

[0074] As illustrated in FIG. 9, the maximum value of the plate thickness reduction rate of the press-formed product **110** having the intermediate shape was 14.8%, and the maximum value of the plate thickness reduction rate of the press-formed product **210** having the target shape was 18.5%. The result was smaller than the maximum value of the plate thickness reduction rate of the press-formed product **210** (=21.5%) obtained by the method of press-forming in one step conventionally illustrated in FIG. 4. Similarly, the result was smaller than the maximum value of the plate

thickness reduction rate of the press-formed product **210** (=25.4%) obtained by the method of Patent Literature 1 illustrated in FIG. 6.

[0075] The results suggest that, according to the press-forming method of the embodiment, it is possible to suppress the occurrence of stretch flange cracking at the concave curved portion **221** in the press-forming process and to press-forming the press-formed product **210** having the target shape.

[0076] As described above, in the press-forming method according to the embodiment, first, the blank **100** (FIG. 8(a)) provided with the concave shape **101** corresponding to the curve of the concave curved portion **221** of the press-formed product **210** having the target shape is formed. Next, in the press-forming method according to the embodiment, the formed blank **100** is press-formed into the press-formed product **110** having the intermediate shape in which the cutout portion **125** is provided at the central portion **121a** of the curve of the concave curved portion **121** (FIG. 9(a)). Further, in the press-forming method according to the embodiment, the press-formed product **110** having the intermediate shape is press-formed into the press-formed product **210** having the target shape (FIG. 9(b)).

[0077] Accordingly, in the process of press-forming the press-formed product **110** having the intermediate shape, the tensile stress acting on the vicinity of the cutout portion **125** at the concave curved portion **121** can be reduced, and the tensile stress can be dispersed to both end sides of the curve at the concave curved portion **121**.

[0078] As a result, when the press-formed product **110** having the intermediate shape is press-formed into the press-formed product **210** having the target shape, stretch flange cracking can be sufficiently suppressed, stable press-forming without cracking can be performed, and productivity and yield can be improved.

[0079] Furthermore, the press-forming method according to the embodiment does not require moving the blank **100** during the press-forming as in the method described in Patent Literature 2. Therefore, according to the press-forming method according to the embodiment, it is possible to obtain a good press-formed product **210** having the target shape while suppressing a variation in dimensional accuracy, and it is possible to improve productivity and yield.

[0080] In the above description, as illustrated in FIG. 10 described above, the ridge portion **113** of the concave curved portion **121** having the intermediate shape is positioned on the outer side of the ridge portion **213** of the concave curved portion **221** having the target shape, but the present invention is not limited thereto.

[0081] However, in the present invention, as illustrated in FIG. 10, the ridge portion **113** of the concave curved portion **121** having the intermediate shape is preferably positioned on the outer side of the ridge portion **213** of the concave curved portion **221** having the target shape. Here, as compared with the concave curved portion **221** having the target shape, a material remains in the top portion **111**, the ridge portion **113**, and the vertical wall portion **115** at the concave curved portion **121**. As a result, even when the vertical wall portion **115** and the flange portion **119** having the intermediate shape are partially cut out, the press-formed product **210** having the target shape including the top portion **211**, the vertical wall portion **215**, and the flange portion **219** can be reliably press-formed.

[0082] Furthermore, when the ridge portion **113** of the concave curved portion **121** having the intermediate shape is positioned on the outer side of the ridge portion **213** having the target shape, it is possible to secure a material that follows a material flow caused by local stretch flange deformation due to the forming of the concave curved portion **121** in the intermediate shape forming Step S3. Then, in the subsequent target shape forming Step S5, strain generated at the end portion of the flange portion **219** of the concave curved portion **221** having the target shape to be press-formed can also be dispersed in a wide range and be reduced, and the stretch flange cracking can be further suppressed.

[0083] In addition, when the ridge portion **113** of the concave curved portion **121** having the

intermediate shape is positioned on the outer side of the ridge portion **213** having the target shape, the concave curved portion **121** may have a constant radius of curvature of curve. Alternatively, as illustrated in FIG. **10**, the concave curved portion **121** having the intermediate shape may include the central portion **121a** of the curve (radius of curvature R.sub.C) and an end portion **121b** of the curve (radius of curvature R.sub.E) having different curvature radii. In addition, the radius of curvature R.sub.E at the end portion **121b** of the curve of the concave curved portion **121** having the intermediate shape is not limited regarding a radius of curvature R.sub.0 at the concave curved portion **221** having the target shape, and any of $R_{sub.0} < R_{sub.E}$ and $R_{sub.0} > R_{sub.E}$ is possible. It may be $R_{sub.0} = R_{sub.E}$.

[0084] Furthermore, in the above case, the length (ridge length) of the ridge portion **113** at the concave curved portion **221** having the intermediate shape may be made longer than the length (ridge length) of the ridge portion **213** at the concave curved portion **221** having the target shape. As a result, it is preferable to secure a larger amount of material that follows the material flow caused by the local stretch flange deformation due to the forming of the concave curved portion **121** in the intermediate shape forming step.

[0085] In the above embodiment, the press-formed product **210** having the Z-shaped cross-sectional shape is to be formed, and as illustrated in (i) of FIG. **11(a)**, the flange portion **119** and the vertical wall portion **115** of the press-formed product **110** having the intermediate shape are partially cut out. However, in the present invention, as illustrated in (ii) of FIG. **11(a)**, the press-formed product **110** having the intermediate shape may be provided with a cutout portion **127** having a shape in which the flange portion **119**, the vertical wall portion **115**, and the top portion **111** at the concave curved portion **121** are partially cut out. As such, the “cutout portion” of the press-formed product **110** having the intermediate shape and includes the flange portion **119** refers to a portion that should originally be the vertical wall portion **115** or the top portion **111** and a portion without any material.

[0086] In addition, as described above, the press-formed product **230** having the backward C-shaped cross-sectional shape (FIG. **2(b)**), the press-formed product having the L-shaped cross-sectional shape, or the press-formed product **250** having the hat cross-sectional shape (FIG. **2(c)**) may be formed in the present invention.

[0087] When the press-formed product having the L-shaped cross-sectional shape is to be formed, first, a press-formed product **130** as illustrated in FIG. **11(b)** is press-formed as an intermediate shape, and the press-formed product is press-formed into a press-formed product having a target shape. Here, as illustrated in FIG. **11(b)**, the press-formed product **130** having the intermediate shape includes a top portion **131** and a vertical wall portion **135** continuous from the top portion **131** with a ridge portion **133** interposed therebetween. The press-formed product **130** having the intermediate shape includes a concave curved portion **141** that is curved in the concave shape in the top view.

[0088] Then, as illustrated in FIG. **11(b)**, press-forming may be performed so that a part of the concave shape **101** provided in the blank **100** (FIG. **8(a)**) becomes a cutout portion **147** at a central portion **141a** of the curve of the concave curved portion **141**. Here, the cutout portion **147** illustrated in FIG. **11(b)** has a shape in which the vertical wall portion **135** and the top portion **131** at the central portion **141a** of the curve of the concave curved portion **141** are partially cut out. As described above, the “cutout portion” in the press-formed product **130** having the intermediate shape in the forming of the press-formed product having the L-shaped cross-sectional shape refers to a portion that should originally be the top portion **131** and a portion without any material.

[0089] When the press-formed product **250** having the hat cross-sectional shape is to be formed, the press-formed product having the intermediate shape may be any press-formed product as long as the press-formed product is obtained by performing press-forming so that a part of the concave shape provided in the blank becomes a cutout portion at the central portion of the curve of the concave curved portion. Here, the cutout portion in the press-formed product having the

intermediate shape may have a shape in which the flange portion and the vertical wall portion provided at the central portion of the curve of the concave curved portion or the flange portion, the vertical wall portion, and the top portion are partially cut out.

[0090] To obtain the press-formed product having the intermediate shape and including such a cutout portion, the radius of curvature of the concave shape **101** provided in the blank **100** may be appropriately adjusted in the blank forming step (refer to FIG. **8(a)**). Specifically, for example, the radius of curvature of the concave shape of the blank may be made smaller than the radius of curvature of the ridge portion of the concave curved portion in the top view of the press-formed product having the target shape (refer to the radius of curvature $R_{sub.0}$ in FIG. **10**).

[0091] Alternatively, a position at which the blank **100** is placed in the intermediate shape forming tool of press forming may be adjusted in the intermediate shape forming step so that the press-formed product having the intermediate shape includes the cutout portion. By executing each step of the above-described press-forming method, a target press-formed product can be produced, and the manufactured press-formed product is sufficiently suppressed from having stretch flange cracking.

EXAMPLE

[0092] An analysis was made to demonstrate the operation effects of the press-forming method and the method of manufacturing the press-formed product according to the present invention, and the analysis will be described below.

[0093] In the present example, the press-formed product **250** having the hat cross-sectional shape illustrated in FIG. **12** was a forming target. Here, in the press-formed product **250**, as illustrated in FIG. **12**, a forming height was set to 50 mm, the radius of curvature of the curve of the concave curved portion **261** was set to 150 mm, and bending R of each of the ridge portion **253** and the die shoulder portion **257** was set to 10 mm. Further, in the press-formed product **250**, the length of the curved portion of the ridge portion **253** at the concave curved portion **261** was set to 236 mm.

[0094] First, the blank **100** provided with the concave shape **101** corresponding to the curve of the concave curved portion **261** was formed (refer to FIG. **13(a)**). Here, the radius of the curve of the concave shape **101** was set to 80 mm.

[0095] Next, as illustrated in FIG. **13(a)**, the blank **100** was press-formed into a press-formed product **150** having the intermediate shape (refer to FIG. **13(b)**) using an intermediate shape forming tool of press forming **31** including an intermediate shape forming die **33**, an intermediate shape forming punch **35**, and a pad **37**. Here, regarding a concave curved portion **161** having the intermediate shape, a case in which a central portion **161a** of the curve is linear and the radius of curvature $R_{sub.E}$ of an end portion **161b** of the curve is set to 80 mm was defined as a first invention example. In the first invention example, the ridge portion length of the concave curved portion **161** having the intermediate shape was set to 273 mm. Further, regarding the concave curved portion **161** having the intermediate shape, a case in which the central portion **161a** of the curve is linear and the radius of curvature R_E of the end portion **161b** of the curve is set to 160 mm was defined as a second invention example. The second invention example has a condition that the radius of curvature $R_{sub.E}$ of the end portion **161b** of the curve of the concave curved portion **161** having the intermediate shape is larger than the radius of curvature $R_{sub.0}$ of the concave curved portion **261** having the target shape. In the second invention example, the ridge portion length of the concave curved portion **161** having the intermediate shape was set to 322 mm. As illustrated in FIG. **13(b)**, the press-formed product **150** having the intermediate shape included the concave curved portion **161** that is curved in the concave shape in the top view and linear portions **163** respectively extending from both ends of the concave curved portion **161**. Further, in the press-formed product **150** having the intermediate shape, a cutout portion **165** was provided at a part of a flange portion **159** and a vertical wall portion **155** in the central portion **161a** of the curve of the concave curved portion **161**.

[0096] Subsequently, as illustrated in FIG. **13(b)**, the press-formed product **150** having the

intermediate shape was press-formed into the press-formed product **250** having the target shape using a target shape forming tool of press forming **41** including a target shape forming die **43**, a target shape forming punch **45**, and a pad **47**.

[0097] In addition, in the present example, as a comparison target, as illustrated in FIG. **14**, a case where the blank **100** is press-formed into the press-formed product **250** having the target shape illustrated in FIG. **12** in one step using a tool of press forming **51** including a die **53**, a punch **55**, and a pad **57** was defined as a conventional example. Further, a case in which the press-formed product **250** having the target shape is press-formed in two steps under the condition using a tool of press forming having the shape illustrated in FIG. **5** described above was defined as a comparative example. In the comparative example, the central portion **181a** of the curve of the concave curved portion **181** having the intermediate shape was linear, and the radius of curvature RE of the end portion **181b** of the curve was set to 80 mm. In the comparative example, the ridge portion length of the concave curved portion **181** having the intermediate shape was set to 211 mm. Then, for each of the conventional example and the comparative example, finite element method analysis was performed for the process of press-forming to determine the plate thickness reduction rate.

[0098] FIG. **15** is a contour diagram of the plate thickness reduction rate of the press-formed product **250** having the target shape in the conventional example. FIG. **16** is a contour diagram of the plate thickness reduction rate of each of the press-formed product **150** having the intermediate shape and the press-formed product **250** having the target shape in the comparative example. FIG. **17** is a contour diagram of the plate thickness reduction rate of each of the press-formed product **150** having the intermediate shape and the press-formed product **250** having the target shape in the first invention example. FIG. **18** is a contour diagram of the plate thickness reduction rate of each of the press-formed product **150** having the intermediate shape and the press-formed product **250** having the target shape in the second invention example. Note that, in the conventional example, since press-forming is performed in one step, only the result of the press-formed product **250** having the target shape is illustrated.

[0099] In the conventional example, as illustrated in FIG. **15**, a portion at which the plate thickness decreased the most was an end portion of the flange portion **259** at the concave curved portion **261**, and the maximum value of the plate thickness reduction rate was 20.5%.

[0100] In the comparative example, as illustrated in FIG. **16**, since a line length of a curve of a ridge portion **193** of a press-formed product **190** having the intermediate shape is short, the plate thickness reduction rate of a central portion **201a** of the curve of a concave curved portion **201** is relatively small. However, the maximum value of the plate thickness reduction rate of end portions **201b** of the curve on both end sides of the concave curved portion **201** was 24.4%, that is significantly increased as compared with the maximum value of the plate thickness reduction rate in the press-formed product **250** having the target shape according to the conventional example (=20.5%). Furthermore, in the press-formed product **250** obtained by press-forming the press-formed product **190** having the intermediate shape into the target shape, the maximum value of the plate thickness reduction rate of the end portion of the curve in the concave curved portion **261** was 25.4%, and a result was that tensile strain was concentrated on the concave curved portion **261** and stretch flange cracking was likely to occur.

[0101] On the other hand, in the first invention example, as illustrated in FIG. **17**, the maximum value of the plate thickness reduction rate in the press-formed product **150** having the intermediate shape was 14.8%, and the plate thickness reduction rate could be reduced by providing the cutout portion **165** as compared with the conventional example and the comparative example. Further, in the first invention example, the maximum value of the plate thickness reduction rate in the press-formed product **250** having the target shape was 18.3%. The plate thickness reduction rate was lower than the maximum value of the plate thickness reduction rate in the press-formed product **250** having the target shape according to the conventional example and the press-formed product **250** having the target shape according to the comparative example.

[0102] Further, in the second invention example, as illustrated in FIG. 18, the maximum value of the plate thickness reduction rate in the press-formed product **150** having the intermediate shape was 13.9%, and the plate thickness reduction rate could be reduced by providing the cutout portion **165** as compared with the conventional example and the comparative example. Further, in the second invention example, the maximum value of the plate thickness reduction rate in the press-formed product **250** having the target shape was 18.8%. The plate thickness reduction rate was lower than the maximum value of the plate thickness reduction rate in the press-formed product **250** having the target shape according to the conventional example and the press-formed product **250** having the target shape according to the comparative example. Further, the plate thickness reduction rate at both ends of the concave curved portion **261** in the second invention example was 14.2% at the maximum, and the plate thickness reduction rate could be further reduced than the maximum value of 16.2% of the plate thickness reduction rate at both end portions of the concave curved portion **261** in the first invention example.

[0103] As described above, according to the press-forming method of the present invention, the radius of curvature $R_{sub.E}$ of the end portion **161b** of the curve of the concave curved portion **161** having the intermediate shape can be set with a condition that the radius of curvature $R_{sub.E}$ is larger than the radius of curvature $R_{sub.0}$ of the concave curved portion **261** having the target shape. Furthermore, by appropriately setting the radius of curvature $R_{sub.E}$ of the end portion **161b** of the curve, it is possible to suppress a decrease in plate thickness at both ends of the concave curved portion **261** having the target shape.

[0104] As described above, it was suggested that, according to the press-forming method of the present invention, it is possible to reduce stretch flange cracking at a concave curved portion in the process of press-forming a press-formed product having a target shape and having a concave curved portion.

INDUSTRIAL APPLICABILITY

[0105] The present invention can provide a press-forming method and a method of manufacturing a press-formed product capable of sufficiently suppressing stretch flange cracking of the press-formed product, the press-formed product having a top portion and a vertical wall portion and including a concave curved portion that is curved in a concave shape in a top view.

REFERENCE SIGNS LIST

[0106] **1** INTERMEDIATE SHAPE FORMING TOOL OF PRESS FORMING [0107] **3** INTERMEDIATE SHAPE FORMING DIE [0108] **3a** DIE SIDE INTERMEDIATE SHAPE CONCAVE CURVED PORTION FORMING PORTION [0109] **3b** DIE SIDE INTERMEDIATE SHAPE LINEAR PORTION FORMING PORTION [0110] **5** INTERMEDIATE SHAPE FORMING PUNCH [0111] **5a** PUNCH SIDE INTERMEDIATE SHAPE CONCAVE CURVED PORTION FORMING PORTION [0112] **5b** PUNCH SIDE INTERMEDIATE SHAPE LINEAR PORTION FORMING PORTION [0113] **7** PAD [0114] **11** TARGET SHAPE FORMING TOOL OF PRESS FORMING [0115] **13** TARGET SHAPE FORMING DIE [0116] **13a** DIE SIDE TARGET SHAPE CONCAVE CURVED PORTION FORMING PORTION [0117] **13b** DIE SIDE TARGET SHAPE LINEAR PORTION FORMING PORTION [0118] **15** TARGET SHAPE FORMING PUNCH [0119] **15a** PUNCH SIDE TARGET SHAPE CONCAVE CURVED PORTION FORMING PORTION [0120] **15b** PUNCH SIDE TARGET SHAPE LINEAR PORTION FORMING PORTION [0121] **17** PAD [0122] **21** TOOL OF PRESS FORMING [0123] **23** DIE [0124] **23a** DIE SIDE CONCAVE CURVED PORTION FORMING PORTION [0125] **25** PUNCH [0126] **25a** PUNCH SIDE CONCAVE CURVED PORTION FORMING PORTION [0127] **27** PAD [0128] **31** INTERMEDIATE SHAPE FORMING TOOL OF PRESS FORMING [0129] **33** INTERMEDIATE SHAPE FORMING DIE [0130] **35** INTERMEDIATE SHAPE FORMING PUNCH [0131] **37** PAD [0132] **41** TARGET SHAPE FORMING TOOL OF PRESS FORMING [0133] **43** TARGET SHAPE FORMING DIE [0134] **45** TARGET SHAPE FORMING PUNCH [0135] **47** PAD [0136] **51** TOOL OF PRESS FORMING [0137] **53** DIE [0138] **55**

PUNCH [0139] 57 PAD [0140] 100 BLANK [0141] 101 CONCAVE SHAPE [0142] 110 PRESS-FORMED PRODUCT [0143] 111 TOP PORTION [0144] 113 RIDGE PORTION [0145] 115 VERTICAL WALL PORTION [0146] 117 DIE SHOULDER PORTION [0147] 119 FLANGE PORTION [0148] 121 CONCAVE CURVED PORTION [0149] 121a CENTRAL PORTION OF CURVE [0150] 121b END PORTION OF CURVE [0151] 123 LINEAR PORTION [0152] 125 CUTOUT PORTION [0153] 127 CUTOUT PORTION [0154] 130 PRESS-FORMED PRODUCT [0155] 131 TOP PORTION [0156] 133 RIDGE PORTION [0157] 135 VERTICAL WALL PORTION [0158] 141 CONCAVE CURVED PORTION [0159] 141a CENTRAL PORTION OF CURVE [0160] 147 CUTOUT PORTION [0161] 150 PRESS-FORMED PRODUCT [0162] 153 RIDGE PORTION [0163] 155 VERTICAL WALL PORTION [0164] 157 DIE SHOULDER PORTION [0165] 159 FLANGE PORTION [0166] 161 CONCAVE CURVED PORTION [0167] 161a CENTRAL PORTION OF CURVE [0168] 161b END PORTION OF CURVE [0169] 163 LINEAR PORTION [0170] 165 CUTOUT PORTION [0171] 170 PRESS-FORMED PRODUCT [0172] 173 RIDGE PORTION [0173] 181 CONCAVE CURVED PORTION [0174] 181a CENTRAL PORTION OF CURVE [0175] 181b END PORTION OF CURVE [0176] 190 PRESS-FORMED PRODUCT [0177] 193 RIDGE PORTION [0178] 201 CONCAVE CURVED PORTION [0179] 201a CENTRAL PORTION OF CURVE [0180] 201b END PORTION OF CURVE [0181] 210 PRESS-FORMED PRODUCT [0182] 211 TOP PORTION [0183] 213 RIDGE PORTION [0184] 215 VERTICAL WALL PORTION [0185] 217 DIE SHOULDER PORTION [0186] 219 FLANGE PORTION [0187] 221 CONCAVE CURVED PORTION [0188] 221a CENTRAL PORTION OF CURVE [0189] 223 LINEAR PORTION [0190] 230 PRESS-FORMED PRODUCT [0191] 231 TOP PORTION [0192] 233 RIDGE PORTION [0193] 233a RIDGE PORTION [0194] 235 VERTICAL WALL PORTION [0195] 235a VERTICAL WALL PORTION [0196] 241 CONCAVE CURVED PORTION [0197] 243 LINEAR PORTION [0198] 250 PRESS-FORMED PRODUCT [0199] 251 TOP PORTION [0200] 253 RIDGE PORTION [0201] 253a RIDGE PORTION [0202] 255 VERTICAL WALL PORTION [0203] 255a VERTICAL WALL PORTION [0204] 257 DIE SHOULDER PORTION [0205] 257a DIE SHOULDER PORTION [0206] 259 FLANGE PORTION [0207] 259a FLANGE PORTION [0208] 261 CONCAVE CURVED PORTION [0209] 263 LINEAR PORTION

Claims

1. A press-forming method of press-forming a press-formed product into a target shape, the target shape including a top portion, a vertical wall portion continuous from the top portion, a flange portion continuous from the vertical wall portion, and a concave curved portion curved in a concave shape in a top view, the press-forming method comprising: a blank forming step of forming a blank provided with a concave shape corresponding to a curve of the concave curved portion of the target shape; an intermediate shape forming step of press-forming the formed blank into a press-formed product having an intermediate shape, the intermediate shape having a top portion, a vertical wall portion continuous from the top portion, a flange portion continuous from the vertical wall portion, a concave curved portion curved in a concave shape in a top view, and a cutout portion located at a part of the flange portion and the vertical wall portion or at a part of the flange portion, the vertical wall portion, and the top portion in a central portion of the curve of the concave curved portion; and a target shape forming step of press-forming the press-formed product having the intermediate shape into the target shape, wherein the press-forming is performed in the intermediate shape forming step such that a part of the concave shape provided in the blank becomes the cutout portion.
2. A press-forming method of press-forming a press-formed product into a target shape, the target shape including a top portion, a vertical wall portion continuous from the top portion, and a concave curved portion curved in a concave shape in a top view, the press-forming method

comprising: a blank forming step of forming a blank provided with a concave shape corresponding to a curve of the concave curved portion; an intermediate shape forming step of press-forming the formed blank into a press-formed product having an intermediate shape, the intermediate shape including a top portion, a vertical wall portion continuous from the top portion, a concave curved portion curved in a concave shape in a top view, and a cutout portion located at a part of the vertical wall portion and the top portion in a central portion of the curve of the concave curved portion; and a target shape forming step of press-forming the press-formed product having the intermediate shape into the target shape, wherein press-forming is performed in the intermediate shape forming step such that a part of the concave shape provided in the blank becomes the cutout portion.

3. The press-forming method according to claim 1, wherein, in the intermediate shape forming step, a radius of curvature $R_{sub.E}$ at an end portion of the curve of the concave curved portion having the intermediate shape is set to be larger than a radius of curvature $R_{sub.0}$ at the concave curved portion having the target shape.

4. A method of manufacturing a press-formed product having a target shape, the target shape including a top portion, a vertical wall portion continuous from the top portion, a flange portion continuous from the vertical wall portion, and a concave curved portion curved in a concave shape in a top view, the method comprising: performing press-forming method according to claim 1.

5. A method of manufacturing a press-formed product having a target shape, the target shape including a top portion, a vertical wall portion continuous from the top portion, and a concave curved portion curved in a concave shape in a top view, the method comprising: performing press-forming method according to claim 2.

6. The method of manufacturing a press-formed product according to claim 4, wherein the press-forming is performed in the intermediate shape forming step such that a radius of curvature $R_{sub.E}$ at an end portion of the curve of the concave curved portion having the intermediate shape is set to be larger than a radius of curvature $R_{sub.0}$ at the concave curved portion having the target shape.

7. The method of manufacturing a press-formed product according to claim 5, wherein the press-forming is performed in the intermediate shape forming step such that a radius of curvature $R_{sub.E}$ at an end portion of the curve of the concave curved portion having the intermediate shape is set to be larger than a radius of curvature $R_{sub.0}$ at the concave curved portion having the target shape.

8. The press-forming method according to claim 2, wherein, in the intermediate shape forming step, a radius of curvature $R_{sub.E}$ at an end portion of the curve of the concave curved portion having the intermediate shape is set to be larger than a radius of curvature $R_{sub.0}$ at the concave curved portion having the target shape.
